DEBOPAMA BASU

M.Tech System-on-Chip Design — IIT Palakkad

Research Project Intern — IIT Kharagpur

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2026	M.Tech System-on-Chip Design(EE & CS)	Indian Institute of Technology Palakkad	8.26 / 10
2021	B.Tech Computer Science and Engineering	Future Institute of Technology, MAKAUT, West Bengal	9.12 / 10
2015	Class 12	Gokhale Memorial Girls' School, Kolkata	81.2%
2013	Class 10	Gokhale Memorial Girls' School, Kolkata	93.4%

SKILLS

Tools: Vivado Xilinx HLS HLx, Synopsys Design Compiler, MatLab, SimuLink, VSCode, Jira, Jenkins, Fortify, Bitbucket, SaltMiner, Spring Tool Suite, MySQL, Postman, Git Bash, LTSpice, IPE, Latex

Languages: System Verilog, Verilog, C, C++ Assembly Language (RISC-V), Linux Shell Scripting, Python, Java

Core Subjects: Computer Architecture of High-Performance Processors, Embedded C, Digital Design, Compiler Design, VLSI Design and architecture, Design Verification, Semiconductor device physics (CMOS Technologies), Data Structure and Algorithms, Operating System, Object Oriented Programming, Discrete Mathematics, DBMS

Databases: MySQL, Oracle 10g

FrontEnd: HTML, Bootstrap, CSS, Angular Node js

Hardware Security: Power based Side channel analysis, cryptography, post quantum cryptography, deep learning based side channel analysis

ACHIEVEMENTS & CERTIFICATIONS

- **GATE 2024 (Computer Science and Engineering):** Qualified with a score of 503 and percentile of 95.64.
- **NPTEL Certifications:**
  - 2021: Computer Networks and Internet Protocols (Completed Successfully)
  - 2025: C based VLSI (Elite + Silver with 80% marks)
  - 2026: GPU Architecture and Programming (ongoing)
- **Won 1st Prize in Bit-by-Bit hackathon 2025:** Won first prize in the hackathon organised by IIT Kharagpur and ISEA, sponsored by CDAC where we had to do side channel analysis and deep learning based side channel attacks to recover secret keys by cryptanalysis. Successfully completed 3 problem statements.
- **Other Certifications:**
  - Microsoft Azure Fundamentals

RESEARCH EXPERIENCE & INTERNSHIPS

Research Intern — Indian Institute of Technology Kharagpur — West Bengal **Jun 2025 - Present**

- Working under the supervision of Prof. Debdeep Mukhopadhyay, SEAL (Secured Embedded Architecture Laboratory), on the broad area of hardware security and post-quantum cryptography.
- Learned to perform side-channel attacks on various cryptographic algorithms.
- Working in an incubated startup in IIT Kgp to design a side-channel resilient hardware security module. Successfully integrated a side-channel attack module on post-quantum crypto algorithms in the tool developed in-house.
- Currently working on side-channel attacks on post-quantum crypto algorithms.
- Worked on deep learning-based side-channel analysis, including extensive hyperparameter exploration and performance evaluation of team-developed models.

Application Development and Maintenance Intern — Cognizant — Kolkata **Jan 2021 - Jun 2021**

- Implemented user interfaces using HTML5, CSS3, JavaScript, and Angular as part of frontend development training.
- Developed backend logic using Java with Spring Framework, Spring MVC, and Spring Boot.
- Wrote and executed unit tests using JUnit to ensure code reliability and maintainability.
- Integrated MySQL databases with backend services using JDBC for data persistence and management.

WORK EXPERIENCE

Junior Software Engineer — Cognizant **Jul 2021 - Jun 2023**

- Worked on a project for a fintech-focused payments firm.
- Vertical: Banking and Financial Services.
- Scope: Vulnerability Remediation - Working on Java Spring Boot Microservices.
- Tools/ Software used: Jira, Jenkins CI/CD pipeline, Fortify, Bitbucket, SaltMiner, Spring Tool Suite, MySQL, Postman, Git Bash.
- Remediated Open Source and Static Vulnerabilities (OWASP Top 10) across microservices, as part of the Application Security team, using Fortify. Identified and eliminated false positives from vulnerability reports.
- Initiating Jenkins CI/CD Pipeline and generating and analyzing Scan Reports via Elastic Search (SaltMiner) related to the latest vulnerabilities introduced.
- Actively worked with the team in migrating microservices from Java 8 to Java 11, making necessary changes, writing and implementing JUnit Test cases in the applications.
- Got Microsoft Azure Fundamentals AZ-900 cloud certified during project.

PROJECTS

32-bit RISC-V Simulator **Jul 2024 - Dec 2024**

- Designed and implemented a 32-bit single-cycle RISC-V processor simulator in Python.
- Modified the simulator for a 5-stage pipe-lined processor with hazard detection and forwarding unit mechanisms.

- Integrated a branch predictor into it to optimize control flow decisions.
- Implemented caching mechanisms (instruction and data cache) to improve memory access efficiency.
- Analyzed simulator performance metrics and validated functionality through benchmark programs.
- Enhanced modularity and scalability to support future architectural features.

Verilog Projects (course work) **Jul 2024 - Jun 2025**

- Developed Verilog-based modules for a traffic light controller (using FSM) and for N-bit Fibonacci generator.
- Developed VLSI architectures for algorithms including CORDIC (both iterative and pipe-lined), GCD (Euclid's algorithm), sum of N squares, sum of N numbers, and Nth Fibonacci number, FIR Filter Implementations.
- Performed RTL design and simulation for synchronous counters, FSMs, and arithmetic circuits.
- Conducted timing analysis and optimized digital designs for power and area efficiency.
- Built a Verilog-based down counter with enable and synchronous reset using D-registers and multiplexers.
- Wrote test-benches for functional verification of VLSI modules.

1-D and 2-D DCT Implementation using Systolic Array **Jan 2025 - Jun 2025**

- Implemented 1-D and 2-D Discrete Cosine Transform using the Systolic Array Implementation

SOC Design Projects – Hardware Accelerators Implementation (course work) **Jan 2025 - Jun 2025**

- Designed and implemented hardware accelerators for 512-point FFT, amplitude normalization, Filtfilt filtering, and image processing (convolution), DC Component Removal from a given Signal.
- Wrote C code for each accelerator and utilized Vivado HLS for simulation, synthesis, RTL co-simulation, and IP generation.
- Built block designs using the HLx workflow in Vivado, performed bit-stream generation, and implemented the design on FPGA hardware via Vivado SDK.
- Compared FPGA outputs with MATLAB results to verify functional accuracy and ensure correctness of hardware implementations.
- Designed and implemented face edge detection and systolic peak detection using MatLab, SimuLink , HDL Coder flow.
- Designed a password verifier module and implemented it on FPGA and compared the actual output with the expected output.

Performance Analysis and Optimization of a Multi-threaded application using Intel VTune profiling tool **Jan 2025 - Jun 2025**

- Started with analyzing a normal tiled Matrix Multiplication Code
- Kept on optimizing it using pthread and then OpenMP and analyzed using VTune
- Used vectorization ans SIMD and hierarchical tiling to obtain a further better performance and speedup.

Website development project for an Online Vehicle Management System (with Cognizant) **Jan 2021 - Jun 2021**

- Front-end: HTML5, CSS3, JavaScript, Angular
- Database: MySQL, JDBC
- Backend: Java, Spring Boot, JUnit and followed Agile Methodology.